FIITJEE

ALL INDIA TEST SERIES

FULL TEST – II

JEE (Main)-2021

TEST DATE: 27-12-2020

ANSWERS, HINTS & SOLUTIONS

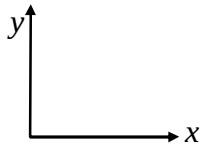
Physics

PART – I

SECTION – A

1. C

Sol. $\vec{V}_w = \frac{v}{\sqrt{2}}\hat{i} + \frac{v}{\sqrt{2}}\hat{j}$



$\vec{V}_m = (at)\hat{j}$

$\vec{V}_{wm} = \frac{v}{\sqrt{2}}\hat{i} + \left(\frac{v}{\sqrt{2}} - at\right)\hat{j}$

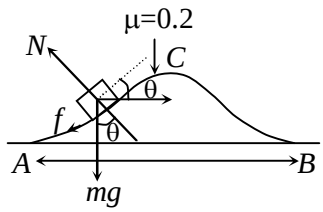
It appears due east when, $\frac{v}{\sqrt{2}} - at = 0$

$\therefore t = \frac{v}{\sqrt{2}a}$

2. C

Sol. Work done by friction = $\int \vec{F} \cdot d\vec{s}$

$= \int_0^x \mu mg\cos\theta \frac{dx}{\cos\theta} = \mu mgx$ = 20 J



3. A

Sol. $T(R-r)=\mu mgR$, $2mg-T=2ma$, $T-\mu mg=ma$

On solving, we get

$\therefore r=R\left(1-\dfrac{3\mu}{2(1+\mu)}\right)$

4. A

Sol. $f=\dfrac{3v}{4(L+0.6r)}$

$\dfrac{df}{dt}=\dfrac{3v}{4}\left(-\dfrac{1}{(L+0.6r)^2}\cdot(0.6)\dfrac{dr}{dt}\right)$

$-2=-\dfrac{3v}{4}\left(0.6\dfrac{dr}{dt}\right)$

$\dfrac{8}{3v\times 0.6}=\dfrac{dr}{dt}$

$\dfrac{dr}{dt}=\dfrac{1}{72}$ m/s

5. C

Sol. $I^2R\,dt=mg\,dx$, $\int_o^t (a_0t)^2R dt=\int_o^h mg\,dx$,

$a_o^2R\int_o^{t_o} dt\,t^2=mgh$, $\dfrac{a_o^2Rt_o^3}{3}=mgh$, $h=\dfrac{a_0^2Rt_0^3}{3mg}$

6. C

Sol. $\dfrac{dQ}{dt}=-KA\dfrac{dT}{dx}$

$\dfrac{dQ}{dt}=-\dfrac{K_0A}{a+bT}\dfrac{dT}{dx}$

$\dfrac{dQ}{dt}\int_0^l dx=-K_0A\int_{T_1}^{T_2}\dfrac{dT}{a+bT}$

$\dfrac{dQ}{dt}=\dfrac{AK_0}{b\,l}\ln\left[\dfrac{a+bT_1}{a+bT_2}\right]$

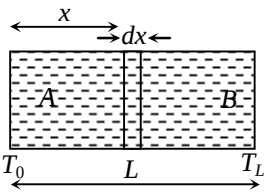
7. A

Sol. $AB\rightarrow$ constant P. T will be increasing with increasing V.
$BC\rightarrow$ constant T. P will be decreasing with increasing V.
$CD\rightarrow$ constant V, decreasing P; hence decreasing T.
$DA\rightarrow$ constant T, decreasing V, increasing P.
Also, BC is at a higher temperature than AD.

8. A

Sol. $v=\sqrt{\frac{\gamma RT}{M}}=\sqrt{\frac{5RT}{3M}}$

$dx=C\cdot dt=\sqrt{\frac{5R}{3M}\left[T_0+\left(\frac{T_L-T_0}{L}\right)x\right]}\,dt, \quad t=\frac{2L}{\sqrt{T_L}+\sqrt{T_0}}\sqrt{\frac{3M}{5R}}$



9. C

Sol. Potential of centre of sphere = $\frac{Kq}{r}+V_i=\frac{Kq}{r}$

where V_i = potential due to induced charge at centre = 0 [$\therefore$ $\Sigma q_i = 0$ and all induced charges are equidistance from centre]

$\therefore$ potential at point $P = \frac{Kq}{r}=\frac{Kq}{r_1}+V_i$ (For conductor all points are equipotential)

$\therefore$ $V_i=K\left(\frac{q}{r}-\frac{q}{r_1}\right)$

10. B

Sol. Potential energy of the system when charge Q is at O is $U_0=\frac{qQ}{a}+\frac{qQ}{a}=\frac{2qQ}{a}$

When charge Q is shifted to position O', the potential energy will be

$U=\frac{qQ}{(a+x)}+\frac{qQ}{(a-x)}=\frac{qQ(2a)}{(a^2-x^2)}=\frac{2qQ}{a}\times\left(1-\frac{x^2}{a^2}\right)^{-1}$

$=\frac{2qQ}{a}\times\left(1+\frac{x^2}{a^2}\right)$ $(\because x << a)$

$\therefore$ $\Delta U = U - U_0 = \frac{2qQ}{a}\left(1+\frac{x^2}{a^2}\right)-\frac{2qQ}{a}=\frac{2qQ}{a^3}(x^2)$

Hence ΔU $\Delta U \propto x^2$.

11. A

Sol. Voltage across rod = $\frac{1}{2}B_0l_0^2\omega_0$

Charge on capacitor = CV_0

$v\times q=\frac{1}{2}CV_0^2+H_{R_1}+H_{R_2}$

$CV_0\times\frac{1}{2}B_0l_0^2\omega_0=\frac{1}{2}CV_0^2+\frac{R_2}{R_1}H_{R_2}+H_{R_2}$

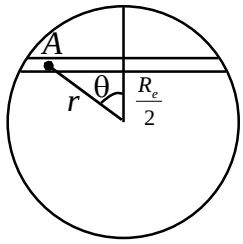
$\frac{1}{2}CV_0B_0l_0^2\omega_0-\frac{1}{2}CV_0^2=\frac{H_{R_2}[R_2+R_1]}{R_1}$

$H_{R_2}=\frac{R_1}{R_1+R_2}\left[\frac{1}{2}CV_0B_0l_0^2\omega_0-\frac{1}{2}CV_0^2\right]=\frac{R_1}{R_1+R_2}\times\frac{1}{2}CV_0[B_0l_0^2\omega_0-V_0]$

$=\frac{1}{2}CV_0^2$

12. D

Sol. Let mass of the earth is M_e and mass of object is m.



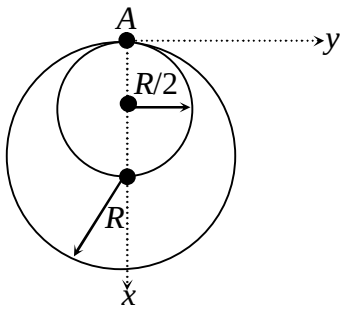
Force on object at $A=\left(\frac{GM_e m}{R^3}r\right)$ towards centre.

Force on object along the tunnel $=\left(\frac{GM_e m}{R^3}r\right)\frac{\sqrt{r^2-\frac{R^2}{4}}}{r}$

Acceleration of object along the tunnel $=\left(\frac{GM_e}{R^3}\right)\sqrt{r^2-\frac{R^2}{4}}$

13. C

Sol. $x_{cm}=\frac{m\left(\frac{R}{2}\right)+2mR}{m+2m}=\frac{5}{6}R$



$I_A=I_1+I_2,\quad I_1=m\left(\frac{R}{2}\right)^2+m\left(\frac{R}{2}\right)^2=\frac{mR^2}{2}$

$I_2=2mR^2+2mR^2=4mR^2\Rightarrow I_A=\frac{9}{2}mR^2$

for compound pendulum $T=2\pi\sqrt{\frac{I}{Mgd}}$

$\Rightarrow$ here M = 3m, $d=\frac{5}{6}R\Rightarrow T=2\pi\sqrt{\frac{9R}{5g}}$

14. A

Sol. Electrostatics force on q = $\frac{\lambda q}{2\pi\varepsilon_0 r}$ away from line charge

Magnetic force = $\frac{\mu_0\lambda v}{2\pi r}\times q\times v$ away from line charge

$\therefore$ total force = $\frac{\lambda q}{2\pi r}\left[\frac{1}{\varepsilon_0}+\mu_0 v^2\right]$

15. D

Sol. $\frac{1}{2}LI_0^{\,2}=\frac{1}{2}(C_1+C_2)V^2$

$V=\left[\frac{LI_0^2}{(C_1+C_2)}\right]^{1/2}$

$Q_1=C_1V=C_1I_0\sqrt{\frac{L}{C_1+C_2}}$

16. C

Sol. $\frac{1}{f}=\frac{1}{v}-\frac{1}{u}, m=\frac{v}{u}$

17. A

Sol. $\Delta x=\left(\mu_1-\mu_2\right) t$

$$I_P=2 I_0\left[1+\cos\left(\frac{2\pi}{\lambda}\right)\left(\mu_1-\mu_2\right) t\right]$$

$$=4 I_0 \cos^2 \frac{\pi}{\lambda}\left(\mu_1-\mu_2\right) t$$

18. A

Sol. $r_n \propto n^2$; $\log \frac{r_n}{r_1}=2 \log n$

19. C

Sol. $C=\frac{\varepsilon_0 A}{L}$

$\therefore \quad \log C=\log \varepsilon_0+\log A-\log L$

$$\frac{dC}{C}=\frac{dA}{A}-\frac{dL}{L}$$

$$\frac{dC}{C}=2\alpha_1 dT-\alpha_2 dT=0$$

$\therefore \quad 2\alpha_1=\alpha_2$

20. C

Sol. $R_{\max}=\frac{u^2}{g} at \ \ \theta=45^0=\left(\frac{mu^2}{2}\right)\left(\frac{2}{mg}\right)=\left(\frac{1}{2} kx^2\right)\left(\frac{2}{mg}\right)$

$\because \quad \left(\frac{1}{2} mu^2=\frac{1}{2} kx^2\right)$

Substituting the values $R_{\max}=\frac{(600)\left(5\times10^{-2}\right)^2}{\left(15\times10^{-3}\right)(10)}=10.0$ m

SECTION – B

21. 3

Sol. $R_{\text{Voltmeter}} = 6\Omega$, $R_{\text{ammeter}} = 0.5\Omega$

$R_{\text{eq}} = 10\Omega$

$I=\frac{30}{10}=3A$

Reading of voltmeter = 1 × 3 = 3 volt.

22. 1

Sol. $F = -\frac{dU}{dr} = -\frac{k}{r}$, $\frac{k}{r} = \frac{mv^2}{r}$ ..... (i)

$E_n = \frac{1}{2}mv^2 + k\ln r$ ..... (ii)

$mvr = \frac{nh}{2\pi}$ ..... (iii)

Solving these $E_n = \frac{k}{2}\left(1+\ln\left(\frac{n^2h^2}{4\pi^2 mk}\right)\right)$

required ratio = $\frac{E_2 - E_1}{E_4 - E_2} = 1.$

SECTION – C

23. 00001.77 (1.75 – 1.80)

Sol. $f \propto \sqrt{g}$

In water $f_w = 0.8\, f_{air}$ $\therefore$ $\frac{g'}{g} = (0.8)^2 = 0.64$

or $\frac{\rho_w}{\rho_m} = 0.36$ ...(i)

In liquid , $\frac{g'}{g} = (0.6)^2 = 0.36$

or $\frac{\rho_L}{\rho_m} = 0.64$

From equations (i) and (ii) $\frac{\rho_L}{\rho_w} = \frac{0.64}{0.36}$

$\therefore S_L = \rho_L/\rho_w = 1.77$

24. 00240.00

Sol. $F_v + F_B = mg$; $F_v + F_B = 3F_B$ $(F_B = \frac{mg}{3})$; $F_B = \frac{F_v}{2}$ = 240 dyne

$F_v = 6\pi\eta rv = 480$ dyne

25. 00006.28

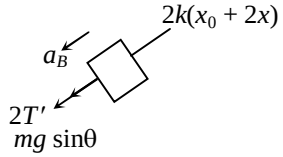
Sol. Let elongation of spring be x_0 in equilibrium. Then,

$$2T + mg\sin\theta = 2kx_0 \quad \ldots(i)$$

And $T = mg$ …(ii)

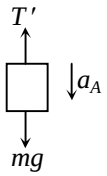
Let Block *B* is displaced by *x* down the inclination

F.B.D. of *B*



$$-ma_B = 2k(x_0 + 2x) - 2T' - mg\sin\theta \quad \ldots(iii)$$

F.B.D. of *A*



$mg - T' = ma_A$

Also, $a_A = 2a_B$

$$T' = mg - 2ma_B$$

$$-ma_B = 2kx_0 + 4kx - 2mg + 4ma_B - mg\sin\theta$$

$$-ma_B = 4kx + 4ma_B$$

$$a_B = -\frac{4k}{5m}x$$

$$\therefore T = 2\pi\sqrt{\frac{5m}{4k}}$$

T = 6.28 s.

Chemistry

PART – II

SECTION – A

26. D
Sol. CO and N_2 both have equal molecular masses and so equal densities.

27. B
Sol. At 8 corners of cube.

28. A
Sol. $KE = h(v - v_0)$

$$\frac{1}{2}mv^2 = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$$

$$v^2 = \frac{2hc}{m}\left(\frac{\lambda_0 - \lambda}{\lambda\lambda_0}\right)$$

29. D
Sol. NO has total 15 electrons and one electron is unpaired.

30. A
Sol. $\Delta n_g = 2 - 4 = -2$

$\Delta n_g RT = -2 \times 8.314 \times 10^{-3} \times 300 = -5.0$ kJ

$\Delta H = -1415 - 5 = 1420$kJ

Heat released for 1.4 g $C_2H_4 = \frac{-1.4 \times 1420}{28}$

$= -71$ kJ

31. A
Sol.

	$2AB_2(g)$	$\rightleftharpoons$	$2AB(g) + B_2(g)$	
at t = 0	2moles(let)	-	-	(x is degree of dissociation)
at eqm.	(2-2x)moles		2xmoles	x mole

Total = 2- 2x = 2x + x = (2 + x) moles;
Total pressure = P

$$K_p = \frac{P^o_{AB} . P_{B_2}}{P^2_{AB_2}}$$

$$\Rightarrow x = \left[\frac{2K_p}{P}\right]^{1/3}$$

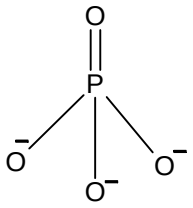
32. B
Sol. $AlCl_3$ dissolves in aqueous NaOH while $FeCl_3$ gives brown ppt. of $Fe(OH)_3$ which on separation can again be converted to respective chlorides.

$$AlCl_3 + 3NaOH \xrightarrow[-3NaCl]{} Al(OH)_3 \xrightarrow{NaOH} \underset{\text{solution}}{Na[Al(OH)_4]}$$

$$FeCl_3 + 3NaOH \xrightarrow[-3NaCl]{} \underset{\text{(brown ppt.)}}{Fe(OH)_3}$$

33. C

Sol.



$$\text{Av. charge} = \frac{-3}{4} = -0.75$$

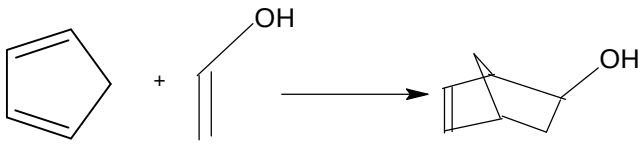
$$\text{Av.B.O} = \frac{2+1+1+1}{4} = 1.25$$

34. A

Sol. Factual

35. A

Sol.



(for 1,4-addition)

36. B

Sol. Factual

37. C

Sol. Lower the gold number, higher is the protective power. Hence, the choice (c).

38. A

Sol. Factual

39. B

Sol. Rate $\propto [CH_4][Cl]$; 2nd order reaction.

40. D

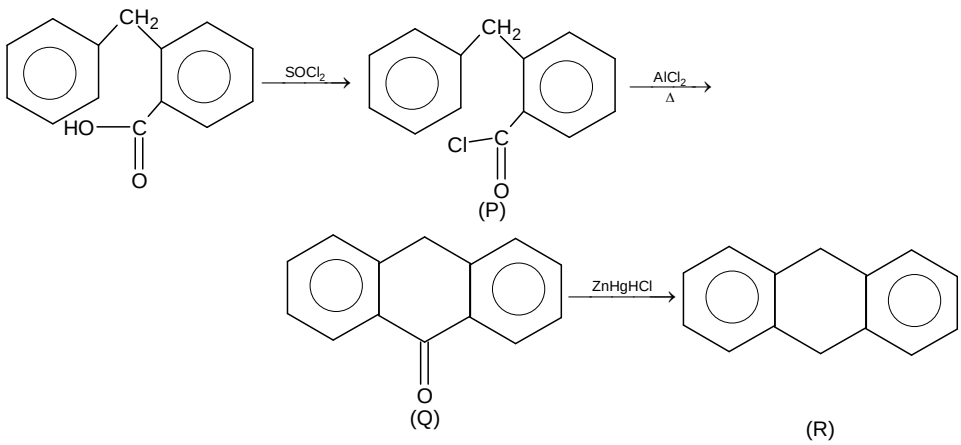
Sol. Factual

41. C

Sol. Factual

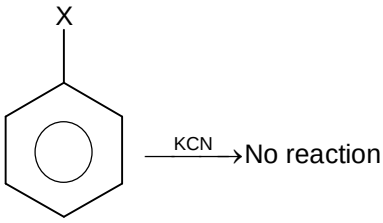
42. C

Sol.



43. A

Sol.



44. C

Sol. Bakelite is not a polyamide., it is phenol formaldehyde resin.

45. C

Sol. $-\overset{\overset{O}{||}}{C}-NH-$ is amide bond which, in peptides, is called peptide bond.

SECTION – B

46. 8

Sol. $A + 3B \rightleftharpoons 2C$

1 mol of A reacts with 3 mol of B.

5 mol of A will react with 15 mol of B.

Given mol of B are 12 which is limiting.

Now, 3 mol of B will give 2 mol of C.

So, 12 mol of B will give $\frac{2\times12}{3}$,i.e., 8 mol of C.

47. 4

Sol. $NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

At t = 0 solid _ _

At eqm. _ 2 moles 1 mole ; n_{total}=3 moles

$K_{kp} = (P_{NH_3})^2(P_{CO_2})$

For $P = 3$ atm

$K_p = \left(\frac{2}{3}\times3\right)^2 \times \left(\frac{1}{3}\times3\right) = 4$

SECTION – C

48. 00500.00

Sol. Let the initial number of moles in vessel be 5 at 27°C, i.e., 300 K. Moles of air left at new temperature

$= = 5 = \frac{2}{5}\times5 = 3$

Pressure and volume remaining the same,

$n_1RT_1 = n_2RT_2$

49. 00002.14

Sol. 1 mol Al, n = 3 (electron exchanged)

$\frac{4}{3}\text{mol Al}, n = \frac{4}{3} \times 3 = 4$

$\Delta G^o = -nFE^0 \Rightarrow E^0 = 2.14V$

50. 00126.00

Sol. 20 mL solution of acid contains = $\frac{0.63}{100} \times 20 = 0.126g$

$E = \frac{W \times 1000}{N'V'(\text{alkali})}$

Molar mass = 126 g

Mathematics

PART – III

SECTION – A

51. D

Sol. For ${}^{10}C_x$ to be defined $x \in I, 0 \le x \le 10$.

$\frac{x}{{}^{10}C_x} \le 1 \Rightarrow x \ne 10$ $\quad [x] = x, \{x\} = 0$ as $x \in I$

$\Rightarrow 5x^2 - 60x + 160 > 0$

$\Rightarrow x^2 - 12x + 32 > 0$

$\Rightarrow x \notin [4,8]$

52. C

Sol. Put $x = r\cos\theta, y = r\sin\theta$

$\Rightarrow 5r^2 - r^2\sin 2\theta = 1 \Rightarrow r^2 = \frac{1}{5 - \sin 2\theta}$

r is maximum, when $5 - \sin 2\theta$ is minimum

$5 - \sin 2\theta\,|_{min} = 4 \Rightarrow r^2\,|_{max} = \frac{1}{4} \Rightarrow r\,|_{max} = \frac{1}{2}$

53. B

Sol. Let $f^{-1}(x) = y \quad \Rightarrow x = f(y) \quad \Rightarrow dx = f^{-1}(y)dx$

$I_2 = \int_a^b f(y)(b-y)f'(y)dy = \int_a^b (b-y)f(y)f'(y)dx$

$= \frac{1}{2}I_1 \quad \Rightarrow \frac{I_1}{I_2} = 2$

$\therefore$ option 'B' is correct

54. B

Sol. $\int_0^{5n}\left[3 + \frac{[x]}{5}\right]dx = \int_0^{5n}\left(3 + \left[\frac{[x]}{5}\right]\right)dx = 15n + \int_0^{5n}\left[\frac{x}{5}\right]dx$

$\left(\because \left[\frac{[x]}{5}\right] = \left[\frac{x}{5}\right]\right)$

55. C

Sol. For $x \in \left[1, \frac{\pi}{2}\right], f(x) = \sqrt{\sin 1}$

$\therefore$ Area $= \int_0^{\pi/2}\sqrt{\sin 1}dx = \sqrt{\sin 1}\left(\frac{\pi}{2} - 1\right)$

56. A

Sol. $(A \cup B) \cap B' = A$

$\Rightarrow A' \cup ((A \cup B) \cap B') = A' \cup A = U$

57. B

Sol. Three non parallel lines are Concurrent

$$\begin{vmatrix}\lambda & 2 & 2\\ 2 & \lambda & 3\\ 3 & 3 & \lambda\end{vmatrix}=0$$

$\Rightarrow \lambda = 2, 3, -5$

But $\lambda = 2$, Ist two lines are parallel.

58. D

Sol. $\frac{x}{a}\cos\frac{\alpha+\beta}{2}+\frac{y}{b}\sin\frac{\alpha+\beta}{2}=\cos\frac{\alpha-\beta}{2}\Rightarrow e=\frac{\cos\frac{\alpha-\beta}{2}}{\cos\frac{\alpha+\beta}{2}}.\frac{2\sin\frac{\alpha+\beta}{2}}{2\sin\frac{\alpha+\beta}{2}}=\frac{\sin\alpha+\sin\beta}{\sin(\alpha+\beta)}$

59. A

Sol. Equation of tangent to $\frac{x^2}{4}+y^2=1$ at $(2\cos\theta, \sin\theta)$ is $x\cos\theta+2y\sin\theta=2$ ....(1)

Equation of chord of contact of $P(h, k)$ to $\frac{x^2}{6}+\frac{y^2}{3}=1$ is $hx + 2ky = 6$ ….(2)

Equation (1) & (2) are identical

$\frac{h}{\cos\theta}=\frac{k}{\sin\theta}=3$

$\Rightarrow h = 3\cos\theta$ and $k = 3\sin\theta$

$\Rightarrow h^2+k^2=9$

$\Rightarrow h^2+k^2=6+3$

$\therefore$ it lies on director circle = angle = $\frac{\pi}{2}$.

60. A

Sol. $\frac{dy}{dx}=0\Rightarrow P(-3a, -a)\Rightarrow$ Tangent is $y + a = 0$

Which is asymptote $\Rightarrow a = 1$

61. B

Sol. Area = area inside square formed $x+y=\pm1, x-y=\pm1$ and outside circle $x^2+y^2=\frac{1}{2}$

62. C

Sol. The line $y=\sqrt{3}x$ can be written as $x=\frac{r}{2}$, $y=\frac{r\sqrt{3}}{2}$. If this line cuts the given curve, then

$\frac{r^4}{16}+\frac{ar^3\sqrt{3}}{8}+\frac{br^2\sqrt{3}}{4}+\frac{cr}{2}+\frac{dr\sqrt{3}}{2}+6=0$.

Therefore OA. OB.OC.OD = $|r_1|\,|r_2|\,|r_3|\,|r_4| = |r_1\, r_2\, r_3\, r_4| = 96$

63. B

Sol. Let roots of $2x^2 + 8x + 2 = 0$ are α, β then roots of $ax^2 + bx + c = 0$ are $\alpha + 1, \beta + 1$ sum of roots = $\alpha + 1 + \beta + 1 = -b/a$

$\Rightarrow \quad -4+2=-\frac{b}{a} \quad \Rightarrow \frac{b}{a}=2$ ....(1)

And product of roots $=(\alpha+1)(\beta+1)=\frac{c}{a}$

$\Rightarrow \frac{c}{a} = -2$ ...(2)

$\{\because \alpha + \beta = -4$ and $\alpha\beta = 1\}$

Adding (1) and (2) gives b + c = 0

64. A

Sol. Let point P is $(2\cos\theta, 2\sin\theta)$

$\Rightarrow$ area = 2 $\times$ area of PAB

$= 2\left[\frac{(S_1)^{3/2}}{2a}\right] = \frac{[4\sin^2 0 - 4(2\cos\theta - 4)]^{3/2}}{1}$

$= 8[1 - \cos^2\theta - 2\cos\theta + 4]^{3/2} = 8\left[6 - (\cos\theta + 1)^2\right]^{3/2}$

$\Rightarrow$ Area $\le 8(6)^{3/2}$

Area $\le 48\sqrt{6}$

65. C

Sol. We know that if $d_i = \frac{x_i - A}{h}$ then $\sigma_x = |h|\sigma_d$.

In this case $-2x_i - 3 = \frac{x_i - 3/2}{-1/2}$

So $h = -\frac{1}{2}$.

Thus $\sigma_d = \frac{1}{|h|}\sigma_x = 2 \times 3.5 = 7$

66. D

Sol. $K = 3! \times 3! \times 1 = 36$

$P = 6! \times 3! + 6! = 5040$

67. C

Sol. Conceptual

68. D

Sol. If $P = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \text{adj. } P = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Also $A^{-1} = \frac{\text{adj. } A}{\det . A}$

$\therefore \begin{bmatrix} 2A+1 & -5 \\ -4 & A \end{bmatrix}^{-1} = \frac{\begin{bmatrix} A & 5 \\ 4 & 2A+1 \end{bmatrix}}{2A^2 + A - 20}$

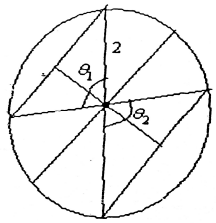
$\therefore \frac{1}{2A^2 + A - 20}\begin{bmatrix} A & 5 \\ 4 & 2A+1 \end{bmatrix}\begin{bmatrix} A-5 & B \\ 2A-2 & C \end{bmatrix} = \begin{bmatrix} 14 & D \\ E & F \end{bmatrix}$

The gives, $\frac{A^2 - 5A - 10}{2A^2 + A - 20} = 14 \qquad \Rightarrow A = 5 \text{ or } -\frac{10}{3}$

$\therefore 3 + \frac{10}{3} = \frac{19}{3}$

69. B

Sol. $\theta_1 = \cos^{-1}\left(\frac{1}{7}\right)$



$\theta_2 = \cos^{-1} 7$

$d_1 + d_2 = 2\cos\frac{\theta_1}{2} + 2\cos\frac{\theta_2}{2}$

$= 4\cos\left(\frac{1}{2}\cos^{-1}\frac{1}{7}\right)$

Let $\cos^{-1}\frac{1}{7} = \theta \Leftrightarrow \frac{1}{7} = \cos\theta$

Then $\cos\left(\frac{1}{2}\cos^{-1}\frac{1}{7}\right) = \cos\frac{\theta}{2}$

$\cos\theta/2 = \frac{2}{\sqrt{7}}$

$\therefore\ d_1 + d_2 = \frac{8}{\sqrt{7}}$

70. A

Sol. Let A denote the event that a sum of 5 occurs, B the event that a sum of 7 occurs and C the event that neither a sum of 5 nor a sum of 7 occurs. We have

$P(A) = \frac{4}{36} = \frac{1}{9}, P(B) = \frac{6}{36} = \frac{1}{6}$ and $P(C) = \frac{26}{36} = \frac{13}{18}$.

Thus,

P (A occurs before B) $= P[A \text{ or } (C \cap A) \text{ or } (C \cap C \cap A) \text{ or } ...]$

$= P(A) + P(C \cap A) + P(C \cap C \cap A) + ...$

$= P(A) + P(C)P(A) + P(C)^2 P(A) + ...$

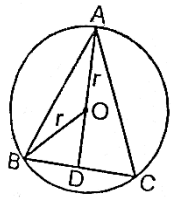
$= \frac{1}{9} + \left(\frac{13}{18}\right)\frac{1}{9} + \left(\frac{13}{18}\right)^2 \frac{1}{9} + ...$

$= \frac{1/9}{1 - 13/18} = \frac{2}{5}$ [sum of an infinite G.P.]

SECTION – B

71. 4

Sol. We have BC = 2BD, AD = h and OD = h – r.



$\therefore BC = 2\sqrt{r^2 - (h-r)^2} = 2\sqrt{2hr - h^2}$

$\Rightarrow AB = \sqrt{2hr - h^2 + h^2} = \sqrt{2hr}$

So that $P = 2AB + BC = 2\left[\sqrt{2hr - h^2} + \sqrt{2hr}\right]$

Also, the area of Δ ABC is

$\Delta = BD \times AD = h\sqrt{2hr - h^2}$

$\therefore \frac{\Delta}{P^3} = \frac{h\sqrt{2hr - h^2}}{8\left(\sqrt{2hr - h^2} + \sqrt{2hr}\right)^3}$; $\frac{\sqrt{2r - h}}{8\left(\sqrt{2r - h} + \sqrt{2r}\right)^3}$

$\Rightarrow \lim_{h \to 0} 512r\frac{\Delta}{P^3} = 512r\frac{\sqrt{2r}}{8(2\sqrt{2}r)^3} = 4$

72. 1

Sol. Clearly $x > 0$. Put $\log_{1/6} x = t$ so that

$|1-t|+2=|3-t| \Rightarrow t \le 1$

$\Rightarrow x \ge 1/6$

SECTION – C

73. 00042.00

Sol. $a+bx+cy = \log 1^2 + \log 2^2 + ... + \log 10^2 - 2\log|\underline{7}$

$\Rightarrow a+bx+cy = 2\left[\log|\underline{10} - \log|\underline{7}\right] = 2\log(10\times 9\times 8)$

$\Rightarrow a+bx+cy = 2\left[1+3\log 2+2\log 3\right] = 2+6x+4y$

$\Rightarrow a=2, b=6, c=4 \Rightarrow 2a+3b+5c = 4+18+20 = 42.$

Hence option 'B' is correct.

74. 00012.24

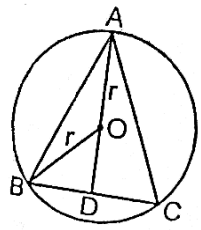
Sol. Since the slope of AP is $\tan^{-1}(-1/x)$ and the slope of BP is $\tan^{-1}(-3/x)$ so

$\theta(x) = \tan^{-1}(-1/x) - \tan^{-1}(-3/x)$

$= \tan^{-1}(3/x) - \tan^{-1}(1/x)$

$= \tan^{-1}(3/x) - \tan^{-1}\dfrac{2x}{x^2+3}$

Since tan x is an increasing function so $\tan\theta(x)$ is maximum when $\theta(x)$ is maximum. But $A = \tan\theta(x) = \dfrac{2x}{x^2+3}$



$\dfrac{dA}{dx} = 2\dfrac{x^2+3-2x^2}{(x^2+3)^2} = 2\dfrac{-x^2+3}{(x^2+3)^2}$

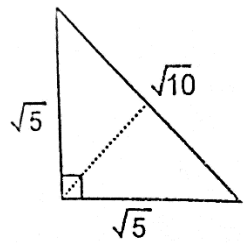
$\dfrac{dA}{dx} = 0 \Rightarrow x = \pm\sqrt{3}.$

Also, $\dfrac{dA}{dx} > 0$ for $x^2 < 3$ and $\dfrac{dA}{dx} < 0$ for $x^2 > 3$. Thus A is maximum when $x = \sqrt{3}$.

Hence max $\theta(x) = \tan^{-1}\dfrac{2\sqrt{3}}{3+3} = \dfrac{\sqrt{3}}{3} = \tan^{-1}1/\sqrt{3} = \pi/6$

75. 00010.24

Sol. $\vec{v}_1 \times \vec{v}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta & -3\sin\theta & -2 \\ 2\cos\theta & -\sin\theta & 1 \end{vmatrix}$



$= -5\sin\theta\hat{i} - 5\cos\theta\hat{j} + 5\sin\theta\cos\theta\hat{k}$

$\therefore$ Area of the $\Delta = \dfrac{1}{2}|\vec{v}_1 \times \vec{v}_2|$

$\dfrac{1}{2}.5\sqrt{1+\sin^2\theta\cos^2\theta} = \dfrac{5}{2}\sqrt{1+\dfrac{1}{4}\sin^2 2\theta}$

For area of Δ to be minimum $\sin 2\theta = 0 \Rightarrow \theta = 0$

Now, $\vec{v}_1 = \hat{i} - 2\hat{k}, \vec{v}_2 = 2\hat{i} + \hat{k}, \vec{v}_3 = \hat{i} + 3\hat{k}$

$|\vec{v}_1| = \sqrt{5}, |\vec{v}_2| = \sqrt{5}, |\vec{v}_3| = \sqrt{10}$

$l = \dfrac{\sqrt{10}}{2} \Rightarrow 2l^2 = 5$